Band-1 Hadamard order 12704 – Classical Paley-I (q=12703); IN SPK via Paley-I; Gap 2052 Still OPEN

H_12704 Paley-I PASS; Paley-II FAIL/SKIP (q=6351=3*29*73 NOT prime; ===3 mod 4 needs ===1); Kronecker FAIL/SKIP (4*3176; 3176===0 mod 4 order-admissible; factors 2^3*397; not_in_seeds / no H_3176); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-04

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. `2026-10-04_Band1_Hadamard_12672_PaleyI`; `2026-10-04_Band1_Hadamard_12660_PaleyI`;
`2026-10-04_Band1_Hadamard_12648_PaleyI`; `2026-10-04_Band1_Hadamard_12620_PaleyI`;
`2026-10-04_Band1_Hadamard_12612_PaleyI`; `2026-10-04_Band1_Hadamard_12584_PaleyI`;
`2026-10-04_Band1_Hadamard_12548_PaleyI`; `2026-10-04_Band1_Hadamard_12540_PaleyI`;
`2026-10-03_Band1_Hadamard_12380_PaleyI`; `2026-10-03_Band1_Hadamard_12348_PaleyI`;
`2026-10-03_Band1_Hadamard_12344_PaleyI`; `2026-10-03_Band1_Hadamard_12324_PaleyI`;
`2026-10-03_Band1_Hadamard_12264_PaleyI`; `2026-10-03_Band1_Hadamard_12252_PaleyI`;
`2026-10-03_Band1_Hadamard_12240_PaleyI`; `2026-10-03_Band1_Hadamard_12228_PaleyI`;
`2026-10-03_Band1_Hadamard_12212_PaleyI`; `2026-10-03_Band1_Hadamard_12204_PaleyI`;
`2026-10-03_Band1_Hadamard_12164_PaleyI`; `2026-10-03_Band1_Hadamard_12144_PaleyI`;
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2026-09-28_Band1_Hadamard_10400_PaleyI; 2026-09-24_Band1_Hadamard_10392_PaleyI;
2026-09-24_Band1_Hadamard_10344_PaleyI; 2026-09-24_Band1_Hadamard_10332_PaleyI;
2026-09-24_Band1_Hadamard_10304_PaleyI; 2026-09-24_Band1_Hadamard_10272_PaleyI;
2026-09-23_Band1_Hadamard_10268_PaleyI; 2026-09-23_Band1_Hadamard_10224_PaleyI;
2026-09-23_Band1_Hadamard_10212_PaleyI; 2026-09-23_Band1_Hadamard_10164_PaleyI;
2026-09-23_Band1_Hadamard_10160_PaleyI; 2026-09-23_Band1_Hadamard_10152_PaleyI;
2026-09-23_Band1_Hadamard_10140_PaleyI; 2026-09-23_Band1_Hadamard_10112_PaleyI;
2026-09-23_Band1_Hadamard_10104_PaleyI; 2026-09-23_Band1_Hadamard_10100_PaleyI;
2026-09-23_Band1_Hadamard_10068_PaleyI.

Lab. /workspace/experiments/band1/hadamard_12704_paleyI/.

Cutoff. 2026-10-04T22:15:58Z (~6:15PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 12704 only (verified max|HHˆT-nI|=0), via classical Paley-I q=12703 (prime ===3 mod 4). 12704 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 12704: Paley-I **PASS** ($q = 12703$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=6351=3*29*73 NOT prime; ===3 mod 4 needs ===1); Kronecker **FAIL/SKIP** (note $12704 = 4 \cdot 3176$; $3176 \equiv 0 \pmod 4$ order-admissible; factors $2^3 \cdot 397$; not_in_seeds / no H_3176). **12704** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **12648+12620+12612+12584+12548+12540+12528+12512+12504+12492+12488+12480+12452+123** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **12704**.

- H_{12704} = Paley-I ($q = 12703$) – classical SPK
- Paley-II for 12704: **FAIL/SKIP** ($q = 6351 = 3 \cdot 29 \cdot 73$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$)
- Kronecker for 12704: **FAIL/SKIP** ($4 \cdot 3176$; $3176 \equiv 0 \pmod 4$ order-admissible; factors $2^3 \cdot 397$; not_in_seeds / no H_3176)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_12704.py`

Results. overall_pass=True; units 488/488; row blocks 100/100 (block size 128); wall 1804.0078s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-04T22:15:58Z (~6:15PM ET)

Arithmetic

- Target $n = 12704 = q + 1$ with $q = 12703$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 6351 = 3 \cdot 29 \cdot 73$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$ — FAIL/SKIP.
- Kronecker: $12704 = 4 \cdot 3176$; $3176 \equiv 0 \pmod 4$ order-admissible; factors $2^3 \cdot 397$; not_in_seeds / no H_3176 — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_12704 Paley-I	Paley-I q=12703	0	1772.4384
H_12704 Paley-II	FAIL/SKIP (q=6351=3*29*73 NOT prime; ===3 mod 4 needs ===1)	–	–
H_12704 Kronecker	FAIL/SKIP (4*3176; 3176===0 mod 4 order-admissible; factors 2ˆ3397; not_in_seeds / no H_3176)	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
12488 in SPK	**True** (Paley-I q=12703)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=12739->n=12740 Paley-I DIRECT (or gap 2052 / Track B)